

Tutorial – 9

1. Design an application which will demonstrate the operations like Insert, Update and Delete on data

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Data.SqlClient;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using System.Xml.Linq;

namespace Tutorial_9
{
    public partial class Form1 : Form
    {
        string connectionString = "Data Source=(localdb)\MSSQLLocalDB;Initial
Catalog=demo;Integrated Security=True;Connect Timeout=30;Encrypt=False;Trusted_Connection
= True";

        public Form1()
        {
            InitializeComponent();

            private void button1_Click(object sender, EventArgs e)
            {
                SqlConnection con = new SqlConnection(connectionString);
                SqlCommand cmd = new SqlCommand("SELECT * FROM student", con);
                SqlDataAdapter da = new SqlDataAdapter(cmd);
                DataTable dt = new DataTable();
                da.Fill(dt);
                dataGridView1.DataSource = dt;
            }

            private void button2_Click(object sender, EventArgs e)
            {
                SqlConnection con = new SqlConnection(connectionString);
```

```
con.Open();

string query = "insert into student values('" + textBox1.Text + "','" + textBox2.Text + "','" +
textBox3.Text + "','" + textBox4.Text + "','" + Convert.ToInt64(textBox5.Text) + "')";

SqlCommand cmd = new SqlCommand(query, con);
cmd.ExecuteNonQuery();

con.Close();
MessageBox.Show("Record Inserted Successfully");
}

private void button3_Click(object sender, EventArgs e)
{
    textBox1.Clear();
    textBox2.Clear();
    textBox3.Clear();
    textBox4.Clear();
    textBox5.Clear();
}

private void button4_Click(object sender, EventArgs e)
{
    if (dataGridView1.CurrentRow == null)
    {
        MessageBox.Show("Please select a row to update!");
        return;
    }

    int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells["id"].Value);

    SqlConnection con = new SqlConnection(connectionString);
    con.Open();

    string query = ("UPDATE student SET " +
        "name = '" + textBox1.Text + "', " +
        "course = '" + textBox2.Text + "', " +
        "dob = '" + textBox3.Text + "', " +
        "address = '" + textBox4.Text + "', " +
        "contact = '" + Convert.ToInt64(textBox5.Text) + "' " +
```

```
        "WHERE id = " + id);
SqlCommand cmd = new SqlCommand(query, con);
cmd.ExecuteNonQuery();
con.Close();

MessageBox.Show("Record Updated Successfully");
}

private void button5_Click(object sender, EventArgs e)
{
    if (dataGridView1.CurrentRow == null)
    {
        MessageBox.Show("Please select a row to delete!");
        return;
    }

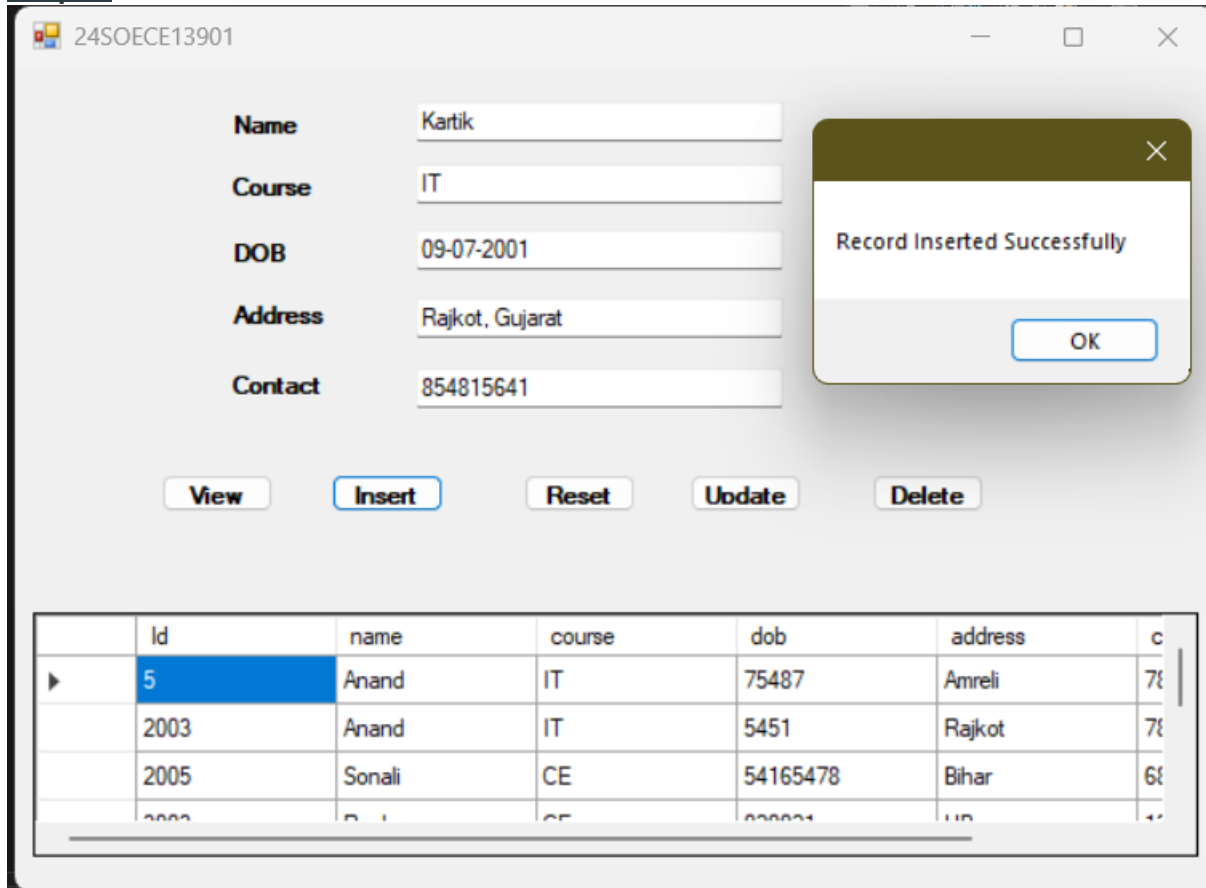
    int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells["id"].Value);

    SqlConnection con = new SqlConnection(connectionString);
    con.Open();
    string query = "DELETE FROM student WHERE id = " + id;
    SqlCommand cmd = new SqlCommand(query, con);
    cmd.ExecuteNonQuery();
    con.Close();

    MessageBox.Show("Record Deleted Successfully");
}

private void dataGridView1_CellContentClick(object sender, DataGridViewCellEventArgs e)
{
    if (e.RowIndex >= 0)
    {
        DataGridViewRow row = dataGridView1.Rows[e.RowIndex];
        textBox1.Text = row.Cells["name"].Value.ToString();
        textBox2.Text = row.Cells["course"].Value.ToString();
        textBox3.Text = row.Cells["dob"].Value.ToString();
        textBox4.Text = row.Cells["address"].Value.ToString();
        textBox5.Text = row.Cells["contact"].Value.ToString();
    }
}
}
```

Output:



	Id	name	course	dob	address	c
▶	5	Anand	IT	75487	Amreli	78
	2003	Anand	IT	5451	Rajkot	78
	2005	Sonali	CE	54165478	Bihar	68
	2003	D. J.	CE	200001	UP	48

2. Design an application which provide the search facility from database records Using DataGridView component.

```

using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Data.SqlClient;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
namespace Tutorial_9
{
    public partial class Form2 : Form

```

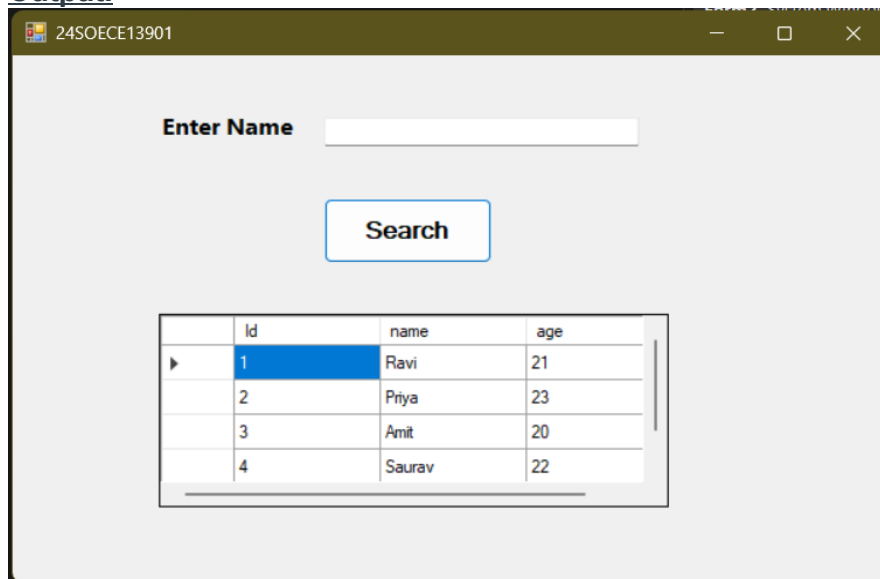
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```
{
    string connectionString = "Data Source=(localdb)\MSSQLLocalDB;Initial
Catalog=demo;Integrated Security=True;Connect
Timeout=30;Encrypt=False;Trusted_Connection = True";
    public Form2()
    {
        InitializeComponent();
    }
    private void button1_Click(object sender, EventArgs e)
    {
        SqlConnection conn = new SqlConnection(connectionString);
        conn.Open();
        SqlCommand cmd = new SqlCommand("SELECT * FROM search WHERE name LIKE
@name", conn);
        cmd.Parameters.AddWithValue("@name", "%" + textBox1.Text + "%");

        SqlDataAdapter da = new SqlDataAdapter(cmd);
        DataTable dt = new DataTable();
        da.Fill(dt);
        dataGridView1.DataSource = dt;
        conn.Close();
    }
}
```

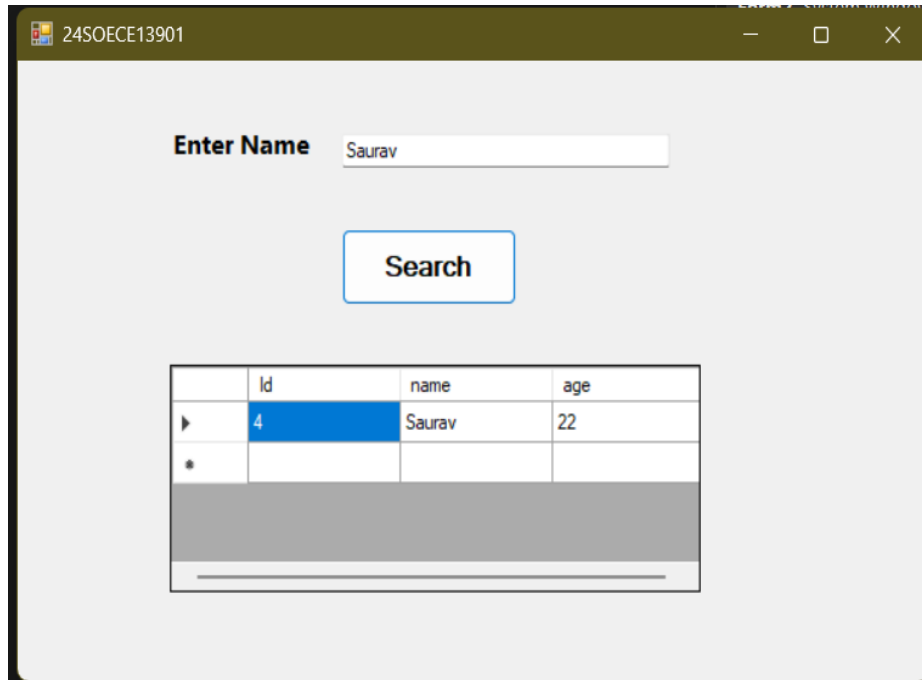
Output:



Id	name	age
1	Ravi	21
2	Priya	23
3	Amit	20
4	Saurav	22

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Enter Name

	Id	name	age
▶	4	Saurav	22
*			